

TO THE EDITOR:

Dreaming of a better future: drug development in myeloma

Richa Parikh^{1,2} and Sagar Lonial^{1,2}¹Department of Hematology and Medical Oncology, Emory University, Atlanta, GA; and ²Winship Cancer Institute of Emory University, Atlanta, GA

"I like the dreams of the future better than the history of the past," a quote by Thomas Jefferson, is an apt summary of the work we have done as a myeloma community to erase the history of the past and move forward with new paradigms. As we consider the "dreams of the future," we need to consider the "nightmares" associated with reemergence of the B-cell maturation antigen (BCMA)-directed antibody-drug conjugate belantamab mafodotin (belamaf). Drug development in multiple myeloma (MM) has been a 20-year fast-paced journey that has yielded many new targets such as CD38, BCMA, and G protein-coupled receptor class C group 5 member D, as well as several next-generation versions of the original "novel agents" such as proteasome inhibitors, immunomodulatory agents, and cereblon E3 ligase modulatory drugs.¹ While some agents have been left behind (panobinostat, programmed cell death protein-1 inhibitors, melflufen)²⁻⁴ after a promising start, few agents have had the up-and-down road that belamaf has had in the past 5 years. Originally, the US Food and Drug Administration approved using the accelerated approval mechanism in 2020 based on the phase 2 DREAMM-2 trial,⁵ there was much excitement over the potential that belamaf could bring for patients with relapsed MM as the first BCMA directed agent approved in MM. Some patients derived significant benefit with long and durable remissions; however, the attempts to mitigate ocular side effects in conjunction with the logistic challenges of collaborating with ophthalmologists, prevented its widespread use. The phase 3 DREAMM-3 study, the confirmatory trial needed for full US Food and Drug Administration approval, demonstrated numerical benefit for the primary end point of progression free survival (PFS) favoring belamaf; however, did not achieve statistical significance⁶ resulting in withdrawal of belamaf from the market in 2023, and it was presumed all was lost.

However, at the same time, 2 large randomized trials had completed enrollment, DREAMM-7 (BVd [belamaf/bortezomib/dexamethasone] vs DVd [daratumumab/bortezomib/dexamethasone]) and DREAMM-8 (BPd [belamaf/pomalidomide/dexamethasone] vs PVd [pomalidomide/bortezomib/dexamethasone]).^{7,8} Both these trials demonstrated a significant improvement in PFS with belamaf compared to the control, with estimated median PFS of 36.6 months in the BVd group vs 13.4 months in the DVd group (hazard ratio for disease progression or death, 0.41; 95% confidence interval, 0.31-0.53; $P < .001$) in the DREAMM-7 trial and median PFS was not reached in the BPd group vs 12.7 months in the PVd group (hazard ratio, 0.52; 95% confidence interval, 0.37-0.73; 2-sided $P < .001$) in the DREAMM-8 trial. These trials reinvigorated the belamaf program and have led to optimism that it may come back and soon be yet another BCMA-directed option for our patients. Where does this resurrected agent fit in the myeloma puzzle? How can we take full advantage of this drug, while minimizing and/or optimally managing toxicity?

First, let us tackle the ocular toxicity that has been front and center in all the trials using belamaf.⁹ Keratopathy is a side effect of the payload mafodotin,¹⁰ which causes microcysts through an impact on corneal stem cells. This is reversible, but at its worst, can cause visual changes and dry/itchy eyes. Belamaf has been typically administered every 3 weeks, with the recommendation to adjust and hold based on ocular toxicity. However, when the actual delivered dose of belamaf was reviewed in DREAMM-7 and -8, the average patient received the drug only every 8 weeks for the first few months, and then every 12 weeks. This was tested in a prospective study where a combination of belamaf with lenalidomide was evaluated in newly diagnosed patients with MM who were transplant-ineligible.¹¹ The

study demonstrated very high overall response rates with lower rates of ocular toxicity when dosing at intervals of every 8 or 12 weeks. Ocular toxicity was assessed using a simple tool in the office eliminating the need for frequent visits to the ophthalmologist, a known logistical challenge for both patients and practitioners. Further, recent post hoc analysis from the DREAMM trials demonstrated similar outcomes for patients receiving every 3 week and less frequent dosing. While perhaps the latter is not quite ready for current adoption, it raises the notion that less frequent drug delivery (in combination with other agents) still maintains efficacy and potentially reduces need for supportive care. These findings need to be confirmed by larger studies in future, but given the proven activity of belamaf combinations in the DREAMM-7 and -8 trials, this could be a convenient option in the relapsed setting. The DREAMM trials, however, included relatively fewer participants from the United States and, as such, their findings need to be contextualized for the US population accordingly. DREAMM-7 (excluded daratumumab refractory patients) presents BVd as an attractive early line option in daratumumab-naïve patients, whereas DREAMM-8 (excluded pomalidomide refractory patients) introduces BPd as an early line option for pomalidomide-naïve patients who are intolerant of bortezomib.

Second, in a world where chimeric antigen receptor T-cells (CAR-T), and bispecific antibodies are the basis for most large trials, how does the myeloma expert and general oncologist know where to position belamaf and how does sequencing impact subsequent therapies? The second part of the question is much harder to answer and really depends on a deeper understanding of the mechanisms of resistance, which after the abrupt withdrawal of belamaf from the market, were not extensively studied. However, it appears that mutations in the receptor or loss of the receptor represent major mechanisms of resistance to BCMA-directed therapies, along with lack of persistence of CAR-T (for BCMA-directed CAR-T).^{12,13} Data sets evaluating sequencing of BCMA-directed agents suggest that prior BCMA-directed therapy limits the efficacy of subsequent therapy, but these are very small cohorts, with a lot of heterogeneity thus, limiting the ability to draw a firm conclusion. However, we do know that binding of a given agent to BCMA, even when a mutation is present, is not necessarily a class effect. While one bispecific may not bind, a CAR-T or a different bispecific may bind, suggesting that this is a complex story. When MM is resistant to belamaf, is it resistant to the payload (chemotherapy resistance) or to immune-mediated killing, or binding of the antibody-drug conjugate overall? Knowing the answer to these questions will help answer questions around sequencing, and will help us to sort out the optimal order of delivery.

In the meantime, not every patient can or wants to receive a CAR-T or a bispecific, given that CAR-T therapy is currently mostly available only at tertiary care centers in selected countries and has unique side effects of cytokine release syndrome, immune-effector cell-associated neurotoxicity syndrome (ICANS), and more recently recognized non-ICANS neurotoxicity and immune-effector cell-associated enterocolitis, while bispecifics require initial hospitalization for step-up dosing to mitigate the risk for cytokine release syndrome and ICANS. Belamaf can be an optimal agent in patients without access to CAR-T or bispecifics or in those who

are not good candidates for both therapies. In contrast, not every patient wants to receive a drug that has a risk of even short-term impact of vision. Patients will vote with their feet about what risks they are willing to take, and we know that the efficacy of belamaf combinations rivals that of any other BCMA-directed therapy in phase 3 trials. In the end, through shared decision-making, we will come up with the approach that makes the most sense for a patient, considering both efficacy and safety standpoints. It is in fact good to know that not all dreams die or are denied, and that some are realized, even if it takes more time.

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ORCID profile: R.P., 0000-0002-5270-104X.

Correspondence: Sagar Lonial, Winship Cancer Institute of Emory University, 1365 Clifton Rd, Building C, Room 4004, Atlanta, GA 30322; email: sloni01@emory.edu.

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